Comparing Dense Retrieval and Re-Ranking

Adobe FY2025 Form 10-K

GitHub Repository

1 Goal and setup

This experiment compares two ways of choosing evidence for a retrieval-augmented generation (RAG) system. The source is Adobe Inc.'s fiscal 2025 Form 10-K, using the extracted text supplied for the project. The pipeline loaded 93 text records/pages and split them into 207 chunks. Chunking used a target of 350 words with 60 words of overlap. Each chunk kept its source page so that evidence could be traced back to the report.

The bi-encoder was `BAAI/bge-base-en-v1.5`, producing a 207 by 768 embedding matrix. Embeddings were normalized, so a vector dot product gives cosine similarity. The cross-encoder was `BAAI/bge-reranker-base`. The local answer model was `Qwen/Qwen2.5-3B-Instruct`. It used the same system and user prompt structure for both pipelines, deterministic decoding, and a limit of 1,024 new tokens. The run used a CUDA environment with two Tesla T4 GPUs available.

2 How the two pipelines differ

Baseline. The bi-encoder creates an embedding for every chunk before any query is asked. At query time, the query is embedded and compared with the stored chunk vectors. The ten highest-scoring chunks are returned, and the top three are sent directly to the answer model.

Re-ranking. It starts with the same ten candidates from dense retrieval. The cross-encoder reads the query and each candidate passage together and gives each pair a relevance score. The candidates are sorted by that score, and the new top three are passed to the same answer model.

The main difference is that a bi-encoder compares separate vectors and is efficient for searching a whole collection, while a cross-encoder evaluates the query and passage together. This gives the cross-encoder a closer look at whether a passage answers the specific question, but it is more expensive to run for every possible query-passage pair. The comparison keeps the original Top 10 fixed so that changes in the final context come from re-ranking, not from running a different first-stage search.

3 Failure-case observations

The rank shifts below use the exported ranking file; a positive shift means that a chunk moved upward. These cases show a more relevant context being selected, but they also show that improved retrieval does not guarantee a fully correct answer.

Case 1 – Generative AI and related risks (Query 1). The baseline selected chunks 10, 19, and 27. Chunk 18, on page 8, was initially at dense rank 8 and moved to re-ranked position 3 (an upward shift of five places). The passage describes Firefly models, training data Adobe has rights to use, Content Credentials, and intellectual-property indemnification for eligible Firefly-powered workflows. The baseline answer said that the excerpts did not disclose specific risks associated with generative AI. The re-ranked answer included more concrete Firefly and IP-indemnification details. This is the clearest example of re-ranking surfacing evidence that was outside the baseline Top 3. The result should still be read carefully: indemnification is a disclosed protection, not a complete inventory of all AI risks. (Form 10-K, p. 8.)

Case 2 – Limits around IP indemnification (Query 7). Chunk 180, which describes customer and channel-partner indemnification, moved from rank 2 to rank 1. Chunk 44, on page 18, moved from rank 7 to rank 3; it discusses risks from third-party AI models, including possible IP-infringement claims and changes to indemnity protections. At the same time, the irrelevant insider-trading-policy chunk 198 fell from rank 3 to rank 9. The baseline answer included an unrelated insider-trading-policy section, while the re-ranked answer replaced that distraction with material more closely related to AI/IP liability. This is a retrieval-level improvement, but not a perfect final answer: the generated response still mixes customer indemnification with separate agreements concerning officers and directors. (Form 10-K, pp. 18 and 80.)

Query	Chunk	Dense → reranked rank	Why the change matters
Q1: generative AI	18	8 → 3	Adds Firefly safeguards and IP-indemnification context.
Q7: IP indemnification	44	7 → 3	Adds related third-party AI/IP liability discussion.
Q7: IP indemnification	198	3 → 9	Pushes an unrelated insider-trading-policy passage down.

Counterexample – legal-proceeding details (Query 6). Re-ranking is not always better. The table-of-contents chunk 3 moved from dense rank 6 to re-ranked rank 1, while the legal-proceeding chunk 183 moved from rank 1 to rank 2 and chunk 184 moved from rank 2 to rank 4. The baseline answer correctly gives January 24, 2024 as the consolidation date for the Shah and Gervat derivative actions; the re-ranked answer gives January 18, 2024. This is a clear reminder to verify dates and legal details against the source, even when the retrieval pipeline runs successfully. (Form 10-K, p. 81.)

4 Latency profiling

The table reports mean per-query measurements from this run. Dense retrieval is shared by both paths. End-to-end time includes retrieval, the cross-encoder step when used, and answer generation; one-time model loading and corpus embedding are not included.

Measured step	Baseline	Re-ranked
Dense retrieval	16.3 ms	16.3 ms
Cross-encoder scoring of Top 10	–	279.6 ms
Answer generation	42.96 s	36.05 s
End-to-end per query	**42.97 s**	**36.35 s**

Dense retrieval averaged about 16 milliseconds, and scoring ten candidates added about 280 milliseconds. Answer generation took much longer and varied noticeably between queries. Although the re-ranked path had a lower mean end-to-end time in this particular run and was faster on six of the ten queries, this should not be interpreted as the cross-encoder making the system faster. The measured generation time varied substantially, and the cross-encoder is an additional step. For a fair timing comparison, the same queries and setup should be timed repeatedly and summarized across runs.

5 Why not use a cross-encoder on the whole collection?

A cross-encoder has to process the query together with each passage. Doing that for every chunk would repeat this more expensive comparison across the full collection for every question. Instead, the bi-encoder cheaply searches the chunk vectors and narrows the search to ten candidates; the cross-encoder then spends its extra work on those candidates. The first stage is built for coverage and speed, while the second stage refines a much smaller list. This becomes increasingly important as the document collection grows.

6 Conclusion and limitations

The outputs contain 100 ranking rows (ten candidates for each of ten queries) and 20 answers, with both answers present for every query and no truncation flags in the generation checks. The most useful observed gains are the promotion of the Firefly passage for Query 1 and the promotion of the AI/IP-risk passage for Query 7. Query 6 shows that re-ranking can also lower useful evidence and affect answer correctness. These are qualitative checks on ten questions, not a formal accuracy score. The generated answers still need source-level review, particularly for numbers, dates, and the difference between customer indemnification and officer/director indemnification.

Source: Adobe Inc., Form 10-K for the fiscal year ended November 28, 2025. Rank shifts and timing figures are taken from this project's exported retrieval and latency results.